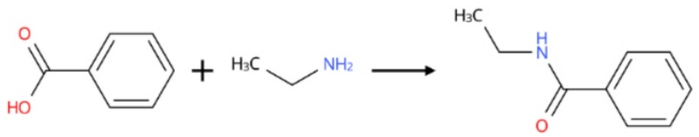
Amidation

AI4-001



№	Reactants/catalysts/reagents	Mol.Wt	Density (g/mL)	Conc. (M)	Equiv.	Amount (mmol)	Volume (mL)	Mass (mg)
1	Benzoic Acid	122.12			1	0.82	0.00	100
2	Ethylamine	45.08			2	1.64	0.00	73.8
3	Diisopropylamine	101.19	0.717		1.4	1.15	0.16	116
4	O-(7-Azabenzotriazol-1-yl)-N,N,N',N'-tetramethyluronium hexafluorophosphate	380.23			1.2	0.98	0.00	374
	Solvents						**Volume (mL)**	
5	Dimethylformamide						1	
	Product	**Mol.Wt**	**Theoretical Yield (mg)**		**Unreacted (mg)**	**Product Mass (mg)**	**% Yield**	
6	N-Ethylbenzamide	149.19	122		5	100	82	

	Hazards	Hazard Rating	Exposure Potential	Risk Rating
1	H315 Causes skin irritation, H318 Causes serious eye damage, H372 Causes damage to organs through prolonged or repeated exposure	VH	L	H
2	H220 Extremely flammable gas, H319 Causes serious eye irritation, H335 May cause respiratory irritation	H	H	H
3	H225 Highly Flammable liquid and vapor, H302 Harmful if swallowed, H314 Causes severe skin burns and eye damage, H332 Harmful if inhaled	H	M	H
4	H228 Flammable solid, H315 Causes skin irritation, H317 May cause an allergic skin reaction, H319 Causes serious eye irritation, H335 May cause respiratory irritation	VH	L	H
5	H226 Flammable liquid and vapor, H312 Harmful in contact with skin, H319 Causes serious eye irritation, H332 Harmful if inhaled, H360D May damage the unborn child	VH	L	H
6	H302 Harmful if swallowed	M	L	L
			Risk Rating:	VH

Sustainability (CHEM21)

Solvents	Safety	Temp °C	Elements	Batch/flow	Isolation	Stoichiometry	Cat. Recovery
Dimethylformamide	VH	20	50-500 years	-select-	-select-	-select-	-select-

Atom Efficiency	Mass Efficiency	Yield	Conversion	Selectivity
23.0	15	82	95	86

Standard Protocols:

- ☐ Cyanide
- ☐ HPLC
- ☐ Mass Spec
- ☐ Pyrophorics
- ☐ Microwave
- ☐ Diazotisation
- ☐ Hydrogenation
- ☐ Peptide Synthesis
- ☐ Ozone
- ☐ Free Radicals
- ☐ Liquid Ammonia
- ☐ Peroxides
- ☐ Sealed Tube

Disposal of Waste Materials:

- ☑ Non-Halogenated Solvent
- ☐ Halogenated Solvent
- ☐ Specialist Container
- ☐ Sink with Excess Water

Spillage Procedure:

- ☑ Standard Spill Response
- ☐ Other Spill Response (Give details on unattended experiment board)

Other Risks, controls, containment, location and PPE.
(Hazardous by Products. Exothermic reactions. The need to inform others of risks. Lab and hood number.)

Hazard categorisation given GLP and other controls specified

Hazard *Potential to cause harm*	◉ 1. Slight ○ 2. Serious ○ 3. Major	**Risk Category (A-D)** ○ **A (10-27)**	**Risk Score HxRxC**
Risk *Likelihood of exposure*	○ 1. Low likelihood ◉ 2. Possible ○ 3. Frequent Occur	○ **B (6-9)**	4
Consequences *Who will be affected*	○ 1. Individual ◉ 2. Local Labs ○ 3. Building wide	◉ **C (3-5)** ○ **D (1-2)**	
Signed:			

Researcher: Joe Davies Supervisor: Super Joe Davies

Benzoic acid, ethylamine, diisopropylamine and HATU were mixed at RT for 2 hours. The reaction mixture was washed with aqueous sodium chloride and extracted in ethyl acetate. The resultant mixture was concentrated under vacuum and purified by silica gel, eluting at 70% ethyl acetate 30% cyclohexane. This yielded an off-white solid (100 mg, 82%).

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